

Exercises

1	2	3
---	---	---

Surname, First name

Machine Learning 1 (52041MAL6Y)

Homework-based Exam 1

17 Sep, 2026, 11:00-13:00

1	1	1	1	1	1	1	1
2	2	2	2	2	2	2	2
3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4
5	5	5	5	5	5	5	5
6	6	6	6	6	6	6	6
7	7	7	7	7	7	7	7
8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9
0	0	0	0	0	0	0	0

a	☒	c	d	e	f	→ b
☒	☒	c	d	e	f	→ a
☒	☒	c	d	e	f	→ b
☒	☒	c	☒	e	f	→ d

Answer multiple-choice questions as shown in the example.

Pay attention to the following:

- **WRITE YOUR NAME AND STUDENT NUMBER ON THE FRONT PAGE** (don't forget to both write **and** mark the digits in the circles)
- Write all your answers in this exam booklet; it will be scanned and digitally graded.
 - Write your **answers inside the boxes** (it is ok, to slightly go over the margins though).
 - The answer boxes should be large enough for your answer. However, **if you need to empty the answer box in order to start over, ask the invigilator for a blank sticker or use the *Emergency Box* at the end of the booklet.**
- You have **2 HOURS** to complete the exam (except for those with pre-approved extensions)
- In total **18 POINTS** can be earned divided over 2 exercises:
 - Exercise 1: Full analysis of a distribution: Exponential distribution [10p]
 - Exercise 2: Linear Regression [8p]
- We work with partial grading: you can get points even if you don't manage to solve the full question. Explaining your procedure clearly may earn you points even if your final calculation contains errors.

During the exam you are allowed to use:

- The provided cheat sheet.
- Basic calculator

Full analysis of a distribution: Exponential distribution

The Poisson process is a model for series of discrete events where an average time between events is known, but the exact time at which they occur is random. It is also assumed that the process is memoryless or Markovian, i.e. the occurrence of a new event is independent of the previous events.

In this exercise, we are interested in analyzing the exponential distribution, which is a continuous probability distribution commonly used to model the time between events in a Poisson process. There are many processes that are modeled this way. For example, it is often used to estimate the lifespan of electronic devices, where failure rates do not change as the device ages. In manufacturing, the time between the completion of successive products on an assembly line can be modeled using an exponential distribution, when the production rate is constant. Even the amount of money customers spend in one trip to the supermarket follows an exponential distribution!

Formally, an exponential distribution with rate parameter $\lambda > 0$ ($X \sim \text{Exp}(\lambda)$) can be modeled as such:

$$f(x | \lambda) = \lambda e^{-\lambda x}, \quad x \geq 0.$$

- 2p **1a** Show that if $X \sim \text{Exp}(\lambda)$, then $\mathbb{E}[X] = \frac{1}{\lambda}$.
Hint: Recall the formula for integration by parts: $\int u dv = uv - \int v du$.

- 1p **1b** Students arrive to the ML1 lecture in the morning according to an exponential distribution with an average time between arrivals of 2 minutes. What is the probability that the time between two consecutive arrivals is less than 1 minute?

Hint: you can give your result both in terms of e or as a numerical approximation (using a calculator).

- 3p **1c** Let $\mathcal{D} = \{x_1, \dots, x_N\}$ be a dataset of independent and identically distributed observations from an exponential distribution with rate λ . Write down the likelihood function $p(\mathcal{D} | \lambda)$ and find the maximum likelihood estimator $\hat{\lambda}_{\text{ML}}$.

↩





3p **1d** Assume a Gamma prior on λ with shape $a > 0$ and rate $b > 0$,

$$p(\lambda \mid a, b) = \frac{b^a}{\Gamma(a)} \lambda^{a-1} e^{-b\lambda}.$$

Use Bayes' rule to write down the posterior distribution $p(\lambda \mid \mathcal{D}, a, b)$ up to a normalizing constant. Then find the maximum a posteriori estimate $\hat{\lambda}_{\text{MAP}}$.





1p **1e** Show that the posterior $p(\lambda \mid \mathcal{D}, a, b)$ follows a Gamma distribution and give its parameters.



Linear Regression

We consider a linear regression model

$$y = \mathbf{x}^T \boldsymbol{\theta},$$

where $\mathbf{x}, \boldsymbol{\theta} \in \mathbb{R}^D$ and $y \in \mathbb{R}$.

We want to fit the model to a dataset $\{(\mathbf{x}_1, y_1), (\mathbf{x}_2, y_2), \dots, (\mathbf{x}_N, y_N)\}$.

The design matrix $X \in \mathbb{R}^{N \times D}$ is formed by stacking the samples row-wise, and the target vector $\mathbf{y} \in \mathbb{R}^N$ contains the corresponding scalar target values. The parameter vector is $\boldsymbol{\theta} \in \mathbb{R}^D$, and the corresponding mean squared error objective is

$$f(\boldsymbol{\theta}) = \frac{1}{N} \|X\boldsymbol{\theta} - \mathbf{y}\|_2^2.$$

2p **2a** Compute the derivative of $f(\boldsymbol{\theta})$ with respect to $\boldsymbol{\theta}$.

- 2p **2b** Find the expression for the value of θ that minimizes the error function f .

- 3p **2c** The least-squares estimator derived above can equivalently be obtained by maximizing the likelihood under a suitable probabilistic model. Introduce such a model and show that maximizing the corresponding likelihood with respect to θ yields the same estimator as in the previous question.
Hint: The p.d.f of the multivariate Gaussian is

$$p(\mathbf{z} \mid \boldsymbol{\mu}, \boldsymbol{\Sigma}) = \frac{1}{(2\pi)^{d/2} |\boldsymbol{\Sigma}|^{1/2}} \exp \left(-\frac{1}{2} (\mathbf{z} - \boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1} (\mathbf{z} - \boldsymbol{\mu}) \right).$$





- 1p **2d** Now suppose we upgrade the linear regression model to a Bayesian framework by placing a prior distribution on the weights θ . After calculating the posterior distribution of θ using the first N datapoints, we start receiving new data one point at a time. Explain how to update the model using the $(N + 1)$ -th datapoint. A formula relating the probability distributions is sufficient; you do not need to perform any explicit calculations.

Note: You no longer have access to the original N datapoints; you only have access to the posterior distribution of the weights after observing the first N datapoints.

Emergency Box

- 3** Use this space only as a last resort to provide answer to a previous question that didn't fit in the corresponding box. Indicate which question it is about (and also indicate it in the answer box of the corresponding question).



